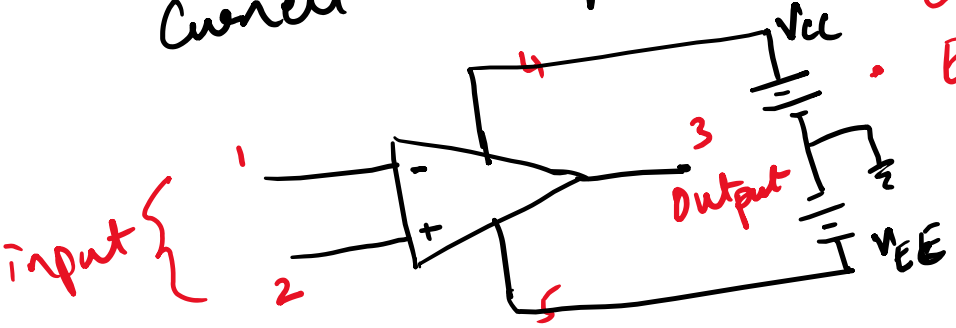
# Operational Amplifiers

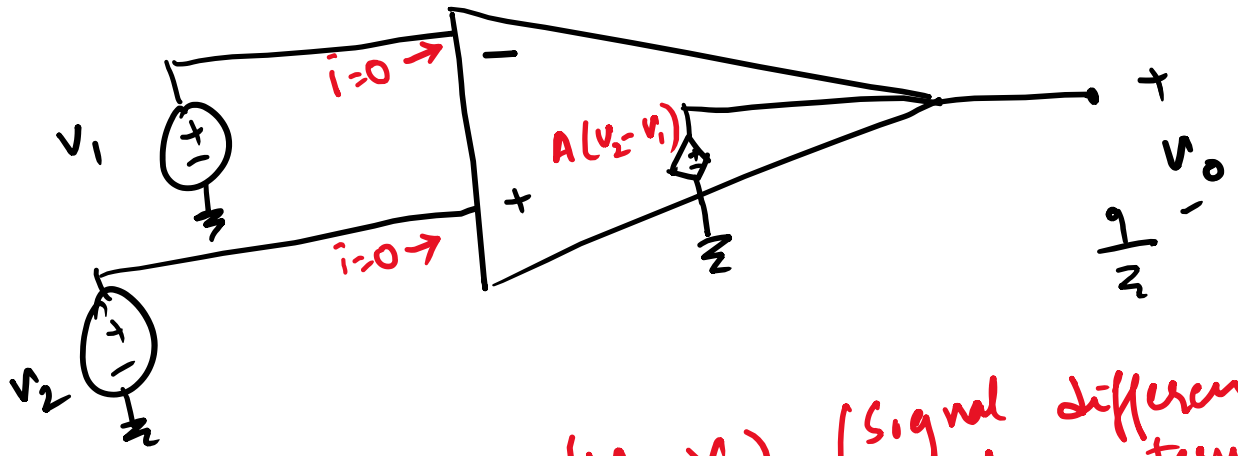
Voltage amp
Current amp.

- Terminal characteristics
- How to analyse circuits containing opamps
- Examples – Summer (Summing amp)
  - Integrator
  - Differentiator




1 : Inverting input
2 : non-inverting input
3 : output

4 : positive power supply $V_{cc}$
5 : negative power supply $V_{EE}$




Opamp senses $(V_2 - V_1)$ (Signal difference between terminals 2 & 1)

$V_0$ or $V_3 = A(V_2 - V_1)$

A : gain of amplification

- Input impedance of opamp is infinite.
- output impedance of opamp is zero.

Differential signal ← Op amp responds to

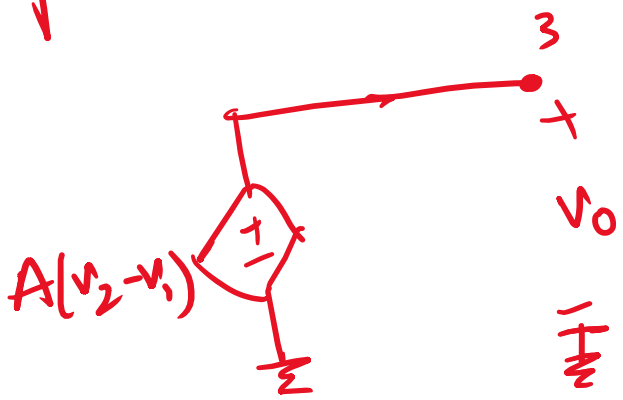
Common signal ← ignores.

Any signal which is common to the inputs is ignored by the opamp.

→ Common Mode Rejection (∞)

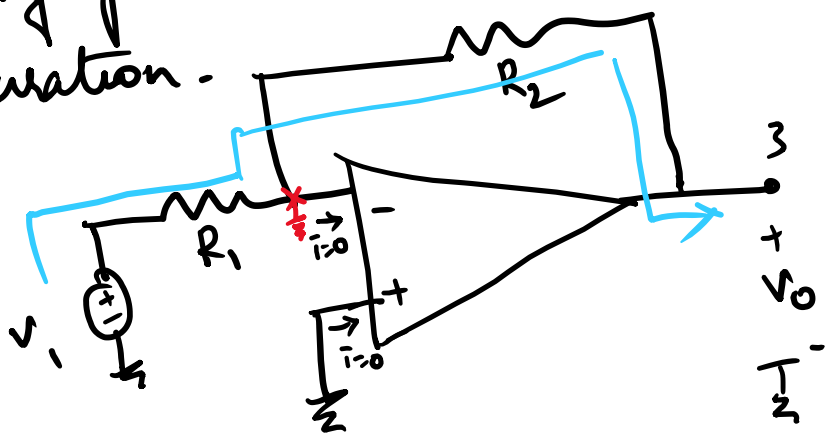
## Open loop gain:

A: Open loop gain.
(differential gain)




## Closed loop configuration.

- Inverting configuration.




•virtual ground: potential at (+)
= potential at (−)

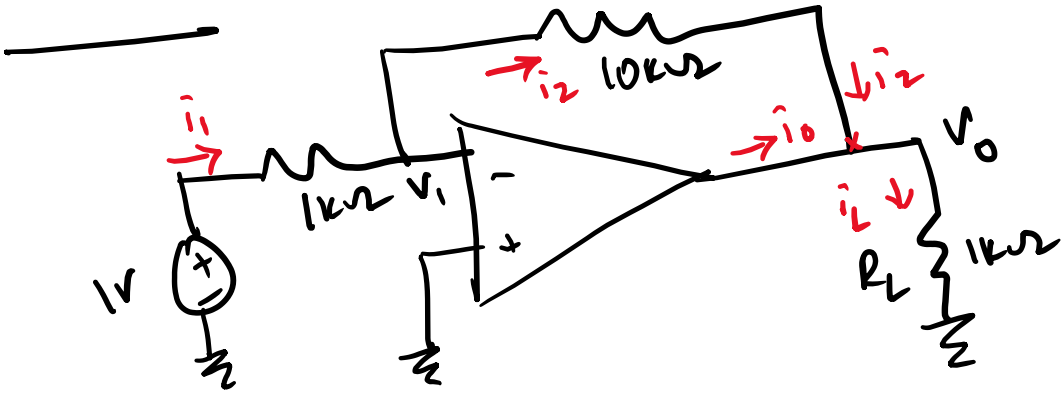
Incoming current at inverting terminal $= \frac{V_1 - 0}{R_1}$

outgoing current $= \frac{0 - V_o}{R_2}$

$$\frac{v_1}{R_1} = -\frac{v_0}{R_2} \quad \Rightarrow \quad v_0 = v_1\left(-\frac{R_2}{R_1}\right)$$

Assume $v_1 = 1V$ $\qquad \frac{R_2}{R_1} = 100$

$v_0 = 100V$ (180° out of phase)




Find $v_1, i_1, i_2, v_o, i_L, i_o, \frac{v_o}{v_i}, \frac{i_L}{i_1}, \frac{P_o}{P_I}$

$v_1 = (v^+) = (v^-) = 0$

$i_1 = \frac{1-0}{1k\Omega} = 1mA = i_2$

$v_o = \left(-\frac{R_2}{R_1}\right)v_i = -10V$ $\qquad i_L = \frac{-10}{1k\Omega} = -10mA$

$i_o + i_2 = i_L \quad \Rightarrow \quad i_o = -11mA$

$P_o = v_o \cdot i_L$

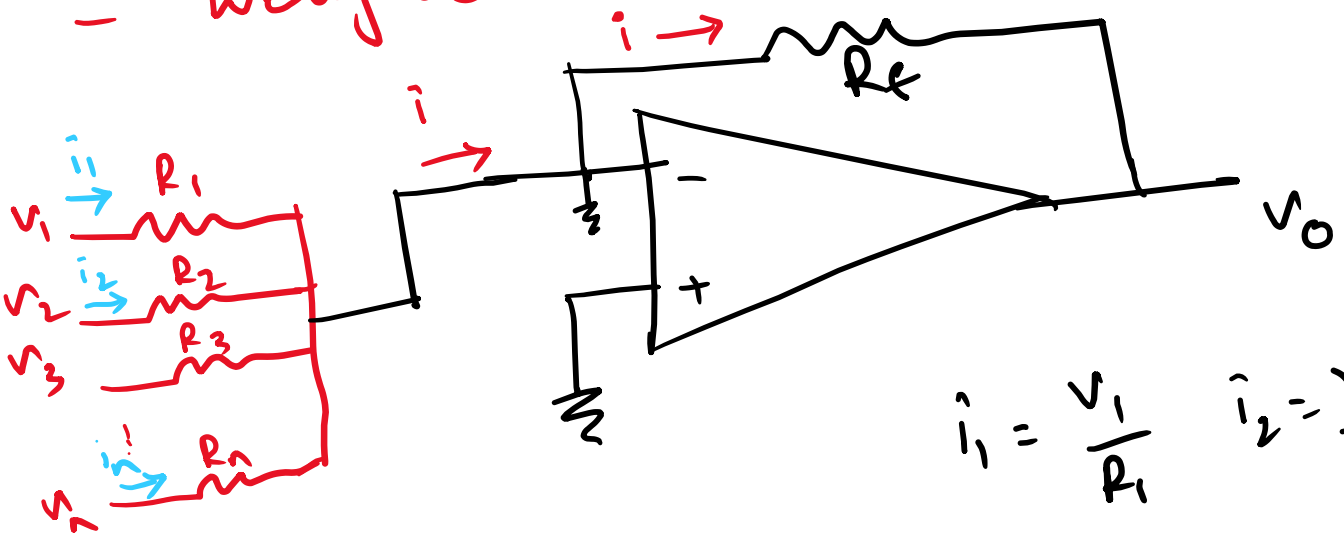
$P_I = v_i \cdot i_1$

$\frac{P_o}{P_I} = \frac{v_o}{v_i} \cdot \frac{i_L}{i_1} = 100W/W$

$\frac{v_o}{v_i} = -10V/V$ $\qquad \frac{i_L}{i_i} = -10\,A/A$

# Application of inverting configuration

## - Weighted Summer.




$$i_1 = \frac{V_1}{R_1} \qquad i_2 = \frac{V_2}{R_2}$$

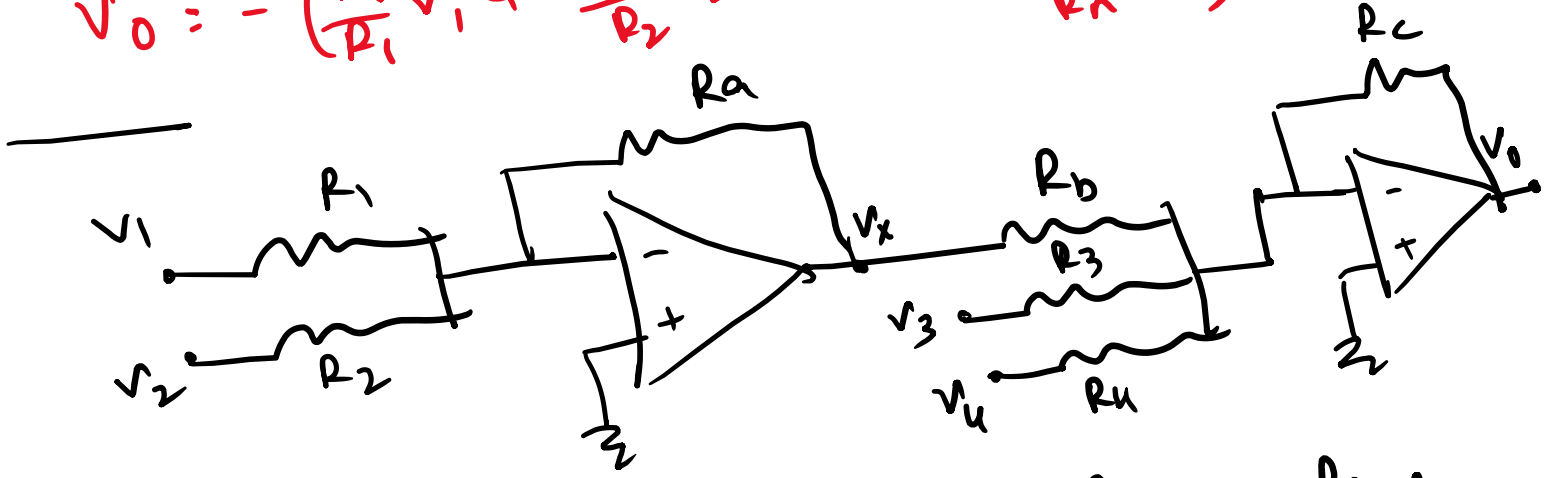
$$i_n = \frac{V_n}{R_n}$$

$$i = i_1 + i_2 + \cdots i_n$$

$$i_1 + i_2 + \cdots i_n = \frac{0 - V_0}{R_f} \Rightarrow V_0 = -i R_f$$

$$V_0 = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \cdots \frac{R_f}{R_n} \cdot V_n\right)$$

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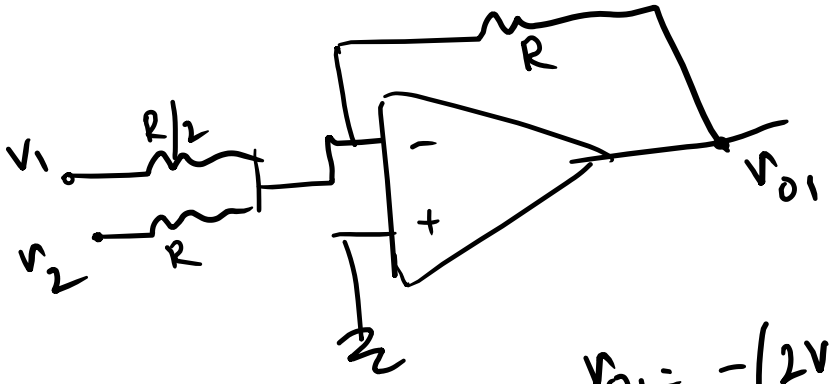

$$V_x = -\left(\frac{R_a}{R_1} \cdot V_1 + \frac{R_a}{R_2} V_2\right)$$

$$V_0 = -\left(\frac{R_C}{R_D} V_x + \frac{R_C}{R_3} V_3 + \frac{R_C}{R_4} V_4\right)$$

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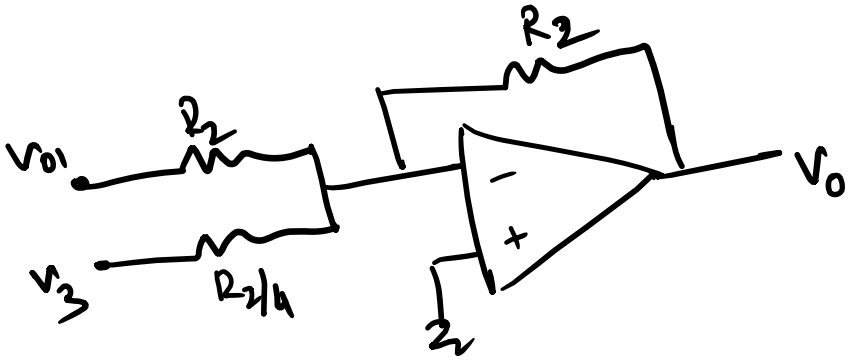
Design a Summer circuit to output

$$V_0 = 2V_1 + V_2 - 4V_3$$




$$-\left(V_1 \cdot \frac{R_f}{R_1} + V_2 \cdot \frac{R_f}{R_2}\right)$$

$$V_{o1} = -(2V_1 + V_2) = \boxed{-2V_1 - V_2}$$




$$V_o = -\left(V_{o1} \cdot \frac{R_2}{R_2} + V_3 \cdot \frac{R_2}{R_2/4}\right)$$

$$= -(-2V_1 - V_2 + 4V_3)$$

$$V_o = 2V_1 + V_2 - 4V_3$$

